⚡ PRODUCTION TEST ARCHITECTURE

Inside a Zero-Flake Playwright + WCAG Starter

Most starter repos give you a folder of sample tests and stop there. Scaling to enterprise CI requires AI guardrails, fast auth, and targeted accessibility gates in one cohesive layout.

🎯 **Core Principle**

When your test selectors already target the browser's Accessibility Tree, fast CI and WCAG 2.1 AA compliance reinforce each other.

AV

Alberto E. Vilano Reyna

Senior SDET & AI Quality Engineer • avilano.dev

5 Layers, One Unified Directory

Every concern (AI rules, diff-based risk selection, fixtures, and accessibility suites) has an explicit boundary:

```
.cursor/rules/          # AI QA guardrails (a11y + POMs)
scripts/risk-engine.ts  # git diff -> targeted tag runner
src/fixtures/           # AxeBuilder + keyboard focus helpers
tests/auth.setup.ts     # <50ms API session state caching
tests/
  ├── a11y/             # Static WCAG 2.1 AA DOM scans
  ├── keyboard/         # Tab order & modal focus traps
  └── screen-reader/    # Guidepup VoiceOver audio checks
```

AV

Alberto E. Vilano Reyna

avilano.dev

Keep CI Under 3 Minutes With a Diff Risk Engine

Running every screen-reader and accessibility suite on documentation edits wastes CI minutes and developer patience.

🧠 **Changed-File Risk Selection**

`scripts/risk-engine.ts` inspects `git diff` on every PR and automatically selects `@a11y` and `@keyboard` suites only when UI components change.

⚡ **Sharded Parallel Execution**

GitHub Actions shards specs across runners so full PR validation completes in under **3 minutes**.

AV

Alberto E. Vilano Reyna
avilano.dev

Sub-50ms API Auth Via storageState

Legacy E2E suites spend up to 40% of their runtime typing credentials into login forms before every spec.

🔑 **Authenticate Once, Reuse Everywhere**

An isolated `auth.setup.ts` fetches a session token via direct API call once and persists `storageState` to disk.

🚀 **Zero UI Login Flakiness**

Every E2E, keyboard, and WCAG spec launches already authenticated in under **50 milliseconds**.

AV

Alberto E. Vilano Reyna

avilano.dev

3 Upgrades for Your Playwright Repo

- ✓ **Pin AI rules in** `.cursor/rules/` before generating new specs.
- ✓ **Cache auth via** `storageState` to cut login runtime to <50ms.
- ✓ **Gate PRs by** `git diff` to run targeted WCAG & keyboard suites in <3 min.

Read the Full Starter Blueprint

Inspect all 5 architectural pillars and the open-source portfolio in the first comment below.

AV
Alberto E. Vilano Reyna
Follow for Playwright & WCAG Architecture • avilano.dev

Save & Follow ★